

OmniCare AI: Enterprise Telehealth & Clinical Telemetry Platform

TECHNICAL SPECIFICATION & RESOURCE ALLOCATION CHARTER

Project ID:	PRJ-HLTH-2026-OMNI	Primary Domain:	Enterprise IT & Cloud
Target Team Size:	4 Core Roles (Scaleable to 6)	Security / Compliance:	HIPAA / HITECH, SOC2 Type II, FHIR R4
Target Deployment:	AWS GovCloud (US-East / US-West)	Engine Target:	PongAI Resource Allocation Engine

1. Executive Summary & Problem Statement

Modern decentralized healthcare systems demand secure, sub-second latency communication between remote physicians and hospital clinical care teams. OmniCare AI delivers an enterprise-grade telehealth and remote physiological telemetry streaming platform. The solution unifies end-to-end encrypted WebRTC peer-to-peer consultations, HL7/FHIR-compliant electronic health record (EHR) integrations, automated clinical triage, and biometric alert dispatching. This project scope outlines the system architecture, required engineering competencies, and team resource allocation matrix for immediate execution in PongAI.

2. Technical Signals & Industry Standards

- WebRTC & P2P Media Pipelines:** Selective Forwarding Unit (SFU) media routing with SRTP/DTLS audio-video encryption.
- FHIR R4 & HL7 Standards:** Standardized patient health data modeling, clinical observation graphs, and secure REST hooks.
- HIPAA & BAA Infrastructure:** AWS GovCloud landing zone, KMS envelope encryption at rest, and audit trail streaming.
- Reactive Clinical Frontend:** React 18 with TypeScript, WebSockets for telemetry gauges, and WCAG 2.1 AA accessibility.
- PostgreSQL & Redis Streams:** Row-Level Security (RLS) persistence, sub-millisecond pub/sub cache, and Timescale metrics.
- Automated Quality Gates:** Playwright end-to-end browser tests, Locust load simulations, and SonarQube static security analysis.

3. Six Core Domain Skill Dimensions

To establish high-confidence candidate matching in PongAI's Player Card radar engine, candidates will be evaluated across six critical domain dimensions:

Skill Dimension	Target Proficiency & Scope
1. WebRTC & Video Streaming	SFU/MCU topologies, adaptive bitrate streaming, ICE/STUN/TURN NAT traversal, SRTP encryption.
2. FHIR & HIPAA Compliance	Health Level Seven (HL7), FHIR R4 schema, BAA compliance, automated audit logging, PHI anonymization.
3. Distributed Data & Caching	PostgreSQL multi-tenant schemas, Redis pub/sub telemetry buffers, time-series clinical metric tables.
4. React & Clinical UI/UX	Real-time medical dashboards, vital signs visualization, accessible telehealth controls, low-latency UI.
5. Cloud & Kubernetes DevOps	AWS EKS container clustering, Terraform IaC, zero-trust network policies, CI/CD pipeline automation.
6. Security & Automated QA	Penetration testing, automated Playwright/Cypress suites, end-to-end regression, defect triage & root cause.

4. Structured Role Breakdown & Actionable Tasks (Team Size: 4)

PongAI's AI decomposition allocates the project scope into four dedicated operational roles with specific actionable deliverables:

Lead Telehealth Architect & Security Officer [Lead]

- Architect end-to-end zero-trust HIPAA security boundaries, TLS 1.3 termination, and KMS master key rotations.
- Design FHIR R4 resource mapping interfaces and clinical access control policies across tenant medical practices.
- Establish architectural governance, code review standards, and technical compliance signoff criteria.
- Conduct threat modeling for WebRTC media streams, patient session replay defenses, and audit log tamper detection.
- Coordinate external SOC2 Type II compliance audit readiness and business associate agreements (BAA).

Real-Time Streaming & Backend Engineer [Specialist]

- Implement scalable WebRTC signaling server with automatic TURN/STUN fallback and session renegotiation.
- Construct asynchronous FastAPI endpoints for patient biometric telemetry ingestion and alert dispatching.
- Configure Redis cluster pub/sub streams to handle 50,000 concurrent physiological vital sign packets per second.
- Develop PostgreSQL schema with Row-Level Security (RLS) isolating electronic medical records by hospital network.
- Integrate Twilio / Agora WebRTC gateway fallbacks for low-bandwidth rural connectivity environments.

Frontend Clinical Portal Developer [Dedicated]

- Build high-responsiveness doctor consultation console in React 18 with picture-in-picture video conferencing.
- Develop patient-facing vitals dashboard featuring real-time Chart.js SVG telemetry graphs for heart rate and SpO2.
- Implement multi-participant screen sharing, clinical whiteboard annotation, and instant document exchange.
- Ensure strict WCAG 2.1 AA accessibility standards, dark mode clinical viewing, and responsive tablet UI.
- Integrate client-side audio/video device selector, connection quality indicators, and automated reconnect prompts.

Cloud Infrastructure & QA Automation Engineer [Specialist]

- Provision reproducible AWS EKS Kubernetes clusters and multi-region failover via Terraform modules.
- Construct automated GitHub Actions CI/CD pipelines deploying ephemeral preview environments for every pull request.
- Author comprehensive Playwright end-to-end integration test suites simulating virtual doctor-patient calls.
- Execute Locust concurrency stress tests simulating 10,000 concurrent patient video sessions with packet loss injection.
- Implement Prometheus and Grafana telemetry dashboards with PagerDuty alerts for latency spikes over 150ms.

5. Phased Delivery Milestones & Acceptance Criteria

Milestone	Target Week	Key Acceptance Deliverables
Phase 1: Foundation	Weeks 1 - 3	HIPAA VPC, EKS cluster, database schemas, and baseline WebRTC signaling.
Phase 2: Consultation MVP	Weeks 4 - 7	Two-way encrypted video consultation, patient portal, and vital telemetry stream.
Phase 3: Integration & EHR	Weeks 8 - 10	FHIR R4 medical record sync, clinical notes, and multi-tenant hospital dashboard.
Phase 4: Hardening & Launch	Weeks 11 - 12	Playwright regression suite, Locust load testing, and penetration security signoff.

Approved for PongAI Resource Ingestion:

Status: Ready for AI Decomposition

Project Leadership: Clinical Technology Steering Committee

Verification Mode: Multipart Form Upload (.pdf)